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Formal Methods in Software Engineering - Assignment #4 (VCC)

1) a) The program is as follows:

```
void main ()
{
    int x = 0;
    int y = 0;
    while (x < 100)
    {
        x = x + 1;
        y = y + 2;
    }
    _assert (y ≥ 102)
}
```

S_1 = `int x = 0; int y = 0;`, b = `x < 100`, S_2 = `x = x + 1; y = y + 2;`

For a loop invariant to be true, it must be valid in three positions:

i) Just before entering the loop

ii) At the start of a new iteration of the loop

iii) Just after the loop is exited

These are the verification conditions for the invariant.

i) $Pre \wedge [S_1] \Rightarrow Inv'$

$\equiv true \wedge [x = 0;\ y = 0;] \Rightarrow x < y$

$\equiv (x = 0 \wedge y = 0) \Rightarrow x < y$

Obviously, because $0 < 0$ is false, the condition is not verified

ii) $Inv \wedge b \wedge [S_2] \Rightarrow Inv'$

$\equiv (x < y) \wedge (x < 100) \wedge [x' = x + 1;\ y' = y + 2] \Rightarrow x' < y'$

$\equiv (x < y) \wedge (x < 100) \wedge [x' = x + 1;\ y' = y + 2] \Rightarrow x + 1 < y + 2$

For all values of $x <$ INT_MAX and $y <$ INT_MAX-1, if $x < y$ holds, $x + 1 < y + 2$ will also hold. Hence this condition is verified.

iii) $Inv \wedge \sim b \wedge [S_3] \Rightarrow Post$

$\equiv (x < y) \wedge \sim(x < 100) \wedge true \Rightarrow y \geq 102$

$\equiv (x < y) \wedge (x \geq 100) \Rightarrow y \geq 102$

$\equiv (100 \leq x) \wedge (x < y) \Rightarrow y \geq 102$

Because a value of $y = 101$ contradicts the above condition, even this condition is not verified.